

N-BED NURSE CALL SYSTEM

System Overview

The N-Bed Nurse Call System is designed to provide reliable communication between patients and nursing staff in a hospital environment. The system consists of multiple ESP32-based patient units connected to a central nurse station through a common Wi-Fi network.

Each patient unit is assigned to a specific room or bed and allows a patient to generate a Nurse Call request. The system also provides Acknowledge and Cancel functions for managing the request. The system supports multiple patient units communicating with a single nurse station. Each request is identified using the corresponding room number and patient information.

The nurse station provides a web-based dashboard that displays active patient requests, request type, call time, acknowledgement time, and current status.

Purpose

The primary purpose of the system is to provide a simple and reliable method for patients to request assistance from nursing staff and for nurses to acknowledge and complete those requests.

The system also records response times and maintains an activity log that can be reviewed through the administration dashboard.

System Components

The system consists of the following major components:

- Multiple ESP32-based patient units
- Central ESP32-based nurse station
- Capacitive touch inputs
- RGB LEDs for visual indication
- Buzzers for audio alerts
- Wi-Fi network
- Nurse station web dashboard
- Hospital administration dashboard

Patient Unit Requirements

Each patient unit shall:

- Use an ESP32 microcontroller.
- Be assigned a unique room or bed number.
- Store the associated patient name.

- Provide a Nurse Call input.
- Provide an Acknowledge input.
- Provide a Cancel input.
- Detect inputs using capacitive touch sensing.
- Provide visual status indication using an RGB LED.
- Provide an audio alert using a buzzer.
- Communicate with the nurse station through Wi-Fi.
- Send the room number and request information when a request is generated.
- Prevent a single continuous button press from generating repeated requests.

Nurse Station Requirements

The nurse station shall:

- Use an ESP32 as the central controller.
- Operate as the central web server.
- Receive requests from multiple patient units.
- Identify requests using their room or bed number.
- Maintain independent request information for different patient units.
- Display multiple active patient requests simultaneously.
- Display the patient name and room number.
- Display the request type.
- Display the call initiation time.
- Display the acknowledgement time.
- Display the current request status.
- Calculate the response time for acknowledged requests.
- Maintain statistics for calls and completed requests.
- Maintain an activity log.

Functional Requirements

Requirement	Description
Request Initialization	The system shall generate a request when the Nurse Call input is activated.
Request Identification	Each request shall contain the corresponding room or bed number and

Requirement	Description
	patient information.
Wireless Communication	Patient units shall communicate with the nurse station through Wi-Fi.
HTTP Communication	Patient units shall transmit requests to the nurse station using HTTP.
Request Display	The nurse station shall display incoming requests on the web dashboard.
Multiple Requests	The nurse station shall support multiple active patient requests.
Independent Requests	Each patient's request shall be managed independently of other patient requests.
Acknowledge Function	The nurse shall be able to acknowledge an active request from the corresponding patient unit.
Cancel Function	The nurse shall be able to cancel a request after the required assistance has been provided.
Call Timestamp	The system shall record the time at which a request is initiated.
Acknowledgement Timestamp	The system shall record the time at which a request is acknowledged.
Response Time	The system shall calculate the time between request initiation and acknowledgement.
Audio Alert	The system shall provide an audio alert when a new request is generated.
Visual Alert	The system shall provide visual indication of different request states.
Activity Logging	The system shall store completed request information in an activity log.
Administration Dashboard	The system shall provide an administration dashboard for viewing system activity.

Request State Management

Each patient request shall follow a defined state sequence:

IDLE → PENDING → ACKNOWLEDGED → COMPLETED

IDLE: No active request is present.

PENDING: A request has been generated and is waiting for nurse attention.

ACKNOWLEDGED: The nurse has acknowledged the request and is attending to the patient.

COMPLETED: The required assistance has been provided and the request has been cancelled.

The state of one patient request shall not affect the state of another patient's request.

Multi-Bed Requirements

The system shall support multiple patient units connected to a single nurse station.

Each patient unit shall have a unique room or bed identifier.

The nurse station shall receive requests from multiple patient units over the same Wi-Fi network.

Requests from different patient units shall appear as separate blocks on the nurse station dashboard.

Each block shall maintain its own patient information, request type, status, call time, and acknowledgement time.

Acknowledging one patient's request shall not acknowledge another patient's request.

Cancelling one patient's request shall not cancel another patient's request.

The system architecture shall allow additional patient units to be added using the same patient-unit firmware with different room and patient information.

Alert and Indication Requirements

The system shall provide both visual and audio feedback.

The RGB LED shall indicate the current state of the patient unit.

The nurse station dashboard shall use color-coded blocks to help nursing staff identify request states.

The buzzer shall provide an audible notification when a new request is generated.

The alert mechanism shall allow nursing staff to notice new requests without continuously monitoring the dashboard.

Time and Response Requirements

The system shall synchronize its clock using an NTP server.

The system shall record the time at which a request is initiated.

The system shall record the time at which the request is acknowledged.

The system shall calculate the response time between the call and acknowledgement.

The response time shall be displayed in seconds for shorter durations and minutes and seconds for longer durations.

The response time shall be stored in the activity log.

Dashboard Requirements

The nurse station dashboard shall display:

- Room or bed number
- Patient name
- Request type
- Request status
- Call time
- Acknowledgement time

- Active request information
The dashboard shall visually differentiate different request states using colors.
The dashboard shall update automatically to reflect changes in request status.

Administration Dashboard Requirements

The system shall provide a password-protected administration dashboard.
The administration dashboard shall display:

- Total calls
 - Active calls
 - Completed calls
 - Medicine requests, where supported
 - Recent activity
 - Room number
 - Request type
 - Call time
 - Acknowledgement time
 - Response time
- The administration dashboard shall allow hospital administration to review request activity and response performance.

Communication Requirements

The patient units and nurse station shall operate on the same Wi-Fi network.
The nurse station shall act as the central server.
Patient units shall send HTTP requests to the nurse station.
Each request shall contain information required to identify the corresponding patient unit and request.
The current prototype shall use HTTP communication without JSON payloads.

Data Storage Requirements

The nurse station shall maintain active request information in ESP32 memory.
The system shall maintain an activity log containing completed request information.
Each log entry shall contain:

- Room number
- Request type
- Call time
- Acknowledgement time

- Response time
The current prototype shall use ESP32 RAM for temporary data storage.
The current prototype shall not require a permanent database.

Administration and Authentication

The administration dashboard shall require authentication before access is granted.
Administrative access shall be separate from the normal nurse station display.
The authentication mechanism shall prevent unauthorized users from accessing administrative information.
For the prototype, administrator credentials are configured within the system.

Reliability Requirements

The system should reliably detect a valid patient input.
A single button press should generate only one request.
The system should maintain the correct request state until the request is acknowledged or cancelled.
The nurse station should correctly associate incoming requests with their corresponding room numbers.
Failure or cancellation of one request should not affect other active requests.

Usability Requirements

The patient unit controls should be simple and easy to operate.
The nurse station dashboard should clearly display important patient and request information.
Request states should be visually distinguishable.
The system should allow nursing staff to identify the room requiring assistance quickly.
The administration dashboard should present activity information in an easily understandable format.

Current Prototype Constraints

The current prototype operates over a local Wi-Fi network.
Patient units currently use the configured network address of the nurse station.
Activity logs are stored in ESP32 RAM.
Stored activity data is lost when the nurse station is restarted.
The current prototype uses HTTP communication without encryption.
The prototype does not currently use a permanent database.
The system requires further security, reliability, and safety validation before deployment in a real hospital environment.

Complete System Flow

Patient request generated → Patient ESP32 detects input → HTTP request sent through Wi-Fi → Nurse station receives request → Request displayed on dashboard → Audio and visual alerts activated → Nurse travels to patient room → Nurse acknowledges request → Acknowledgement time recorded → Response time calculated → Nurse provides assistance → Nurse cancels request → Request marked completed → Activity log updated → System returns to idle state.

